

Sentiment, Emotion, and Topic Analysis of Public Reactions to Presidential Iran-War Communications on Truth Social

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Abstract—This report presents a large-scale natural language processing (NLP) analysis of public reactions on the Truth Social platform to a series of posts made by President Donald J. Trump concerning the 2026 Iran conflict. A corpus of 328,968 cleaned public comments and posts, collected between February 28, 2026 and April 20, 2026, was analyzed using a multi-model sentiment pipeline (VADER, TextBlob, and a Twitter-trained RoBERTa transformer), NRC lexicon-based emotion detection, TF-IDF keyword and n -gram extraction, named entity recognition (NER), Latent Dirichlet Allocation (LDA) and BERTopic topic modeling, aspect-based sentiment analysis, temporal trend analysis, and engagement correlation analysis. Results indicate a polarized but slightly negative-leaning discourse (VADER mean compound score -0.028), with *trust* (28.5%) and *anger* (20.2%) as the two dominant detected emotions. Aspect-level analysis reveals that discourse around “Trump Leadership” skews positive ($+0.115$), while “War/Conflict” (-0.345) and “Media” (-0.225) mentions skew strongly negative. Topic modeling surfaces eight coherent discourse clusters spanning military escalation, diplomacy, religious/patriotic rhetoric, and meme-driven commentary. Engagement analysis shows that positively-toned posts receive, on average, $3\times$ the engagement of neutral posts. These findings offer a quantitative characterization of public sentiment dynamics surrounding a major geopolitical event as expressed on a partisan social media platform.

Index Terms—sentiment analysis, natural language processing, social media analytics, topic modeling, emotion detection, VADER, BERTopic, LDA, aspect-based sentiment analysis, Truth Social

I. INTRODUCTION

Social media platforms have become a primary channel through which political leaders communicate directly with the public, bypassing traditional media intermediaries. Truth Social, launched in 2022, serves as the principal communication channel for President Donald J. Trump. In early 2026, a series of posts referencing military action against Iran generated substantial public reaction on the platform, providing a rich, naturalistic dataset for studying real-time public sentiment toward a fast-evolving geopolitical crisis.

Understanding how such audiences emotionally and rhetorically respond to war-related political communication is of interest to computational social science, political communi-

cation research, and platform governance. This report applies a comprehensive natural language processing pipeline to a corpus of public replies and posts in order to answer the following research questions:

- RQ1** What is the overall sentiment polarity of public reaction, and how consistent are different sentiment models in measuring it?
- RQ2** What discrete emotions (e.g., anger, trust, fear) characterize the discourse?
- RQ3** What topics and entities dominate the conversation, and how does sentiment vary by discourse aspect (e.g., military, diplomacy, media)?
- RQ4** How does sentiment evolve over time, and does emotional tone correlate with user engagement?

II. RELATED WORK

Lexicon-based sentiment tools such as VADER [1] and TextBlob [2] remain widely used for short, informal social media text due to their speed and interpretability, while transformer-based classifiers fine-tuned on Twitter-domain data, such as the CardiffNLP RoBERTa sentiment model [3], offer improved contextual accuracy at higher computational cost. Discrete emotion detection is commonly performed using the NRC Word-Emotion Association Lexicon [4], which maps words to eight basic emotions plus positive/negative polarity. For thematic structure discovery, classical probabilistic topic models such as Latent Dirichlet Allocation (LDA) [5] are often complemented by embedding-based approaches such as BERTopic [6], which clusters document embeddings rather than relying on bag-of-words co-occurrence alone. Named entity recognition was performed using the industrial-strength spaCy pipeline [7]. This report combines these complementary methods into a single analytical pipeline applied to a novel, high-volume political social media corpus.

III. DATASET AND METHODOLOGY

A. Data Source

The dataset was obtained from a publicly hosted Kaggle repository containing scraped Truth Social posts and

comments related to President Trump’s Iran-related communications [8]. The raw corpus consisted of two files: a `posts.csv` file (97 original posts) and a `comments.csv` file (377,624 public replies), merged into a single working dataset of 377,721 records across 50 metadata fields, including engagement counts (replies, reblogs, favourites), account metadata (follower count, verification status), media flags, and timestamps.

B. Preprocessing

Records with missing or empty text content were removed. Text was cleaned by stripping URLs, @-mentions, HTML tags, and non-ASCII characters, and by normalizing whitespace; hashtag symbols were retained as plain words. A lowercase, punctuation-stripped variant was additionally produced for keyword and topic modeling tasks. Timestamps were parsed to derive per-post date, hour-of-day, and weekday features. After cleaning, **328,968** posts remained, with an average of 30.5 words per post (median 16.0). Of these, 2.2% originated from verified accounts and 9.1% contained media attachments. English was the dominant language (96.7% of posts).

C. Sentiment Analysis

Three complementary sentiment models were applied to the cleaned text:

- **VADER** [1]: a rule-based model tuned for social media text, applied to the full corpus. Posts were labeled Positive (compound ≥ 0.05), Negative (compound ≤ -0.05), or Neutral otherwise.
- **TextBlob** [2]: a pattern-based polarity/subjectivity model, applied to the full corpus using the same threshold convention.
- **RoBERTa (Twitter-trained)** [3]: the `cardiffnlp/twitter-roberta-base-sentiment-latest` transformer, applied to a random sample of 5,000 posts due to computational cost, producing three-class (positive/neutral/negative) predictions with associated confidence scores.

D. Emotion Detection

Discrete emotion scores (anger, fear, anticipation, trust, surprise, sadness, disgust, joy) were computed for each post using the NRC Emotion Lexicon [4] via the `nrclex` library. Raw lexicon hit counts were normalized per post to produce a relative emotion-intensity profile, and the arg-max emotion was assigned as the post’s dominant emotion. Posts with zero lexicon matches (16.6% of the corpus) were excluded from the dominant-emotion distribution rather than being forced into an arbitrary class.

E. Keyword, N-gram, and Word Cloud Analysis

Term Frequency–Inverse Document Frequency (TF-IDF) scoring was used to identify salient unigrams, and raw frequency counts were used to extract the most common bigrams and trigrams, after removing English stopwords and platform-specific noise tokens (e.g., “trump,” “rt,” “amp”). Word clouds

were additionally generated for the overall corpus and for each VADER sentiment subgroup to visually contrast vocabulary usage.

F. Named Entity Recognition

Named entities (PERSON, ORG, GPE, LOC, EVENT, LAW, NORP) were extracted from a random sample of 3,000 posts using spaCy’s `en_core_web_sm` pipeline [7], with each document capped at 1,000 characters for processing efficiency.

G. Topic Modeling

Two complementary topic modeling approaches were used. **LDA** [5] was fit with 8 topics on a document-term matrix (2,000-word vocabulary, min-df= 5, max-df= 0.85) using online variational inference. **BERTopic** [6], which clusters sentence-embedding representations and extracts class-based TF-IDF keywords per cluster, was additionally fit on a 5,000-post sample to provide a semantically-driven cross-check of the LDA topics.

H. Aspect-Based Sentiment Analysis

Ten discourse aspects relevant to the Iran conflict (e.g., Iran Military, US Military, Diplomacy, War/Conflict, Economy, Trump Leadership, Media, Israel, National Security, Biden/Democrats) were defined via curated keyword sets. Each post was tagged with all aspects whose keywords it contained, and the post’s VADER compound score was attributed to every matched aspect, yielding an aspect-level sentiment distribution.

I. Temporal and Engagement Analysis

Daily post volume and 3-day rolling-average sentiment were tracked across the full observation window. Engagement was defined as the sum of replies, reblogs, and favourites; its correlation with sentiment, subjectivity, and post length was computed, along with mean engagement stratified by VADER sentiment label.

IV. RESULTS

A. Dataset Overview

Fig. 1 summarizes the corpus composition. Post length is heavily right-skewed, with a median of 16 words; posting volume shows clear day-to-day bursts corresponding to major news events; engagement metrics (replies, reblogs, favourites) are extremely long-tailed, consistent with typical social media engagement distributions; and posting activity is roughly evenly distributed across weekdays.

B. Sentiment Analysis

Table I summarizes the three-way sentiment classification produced by each model. VADER and RoBERTa broadly agree that the discourse leans more positive-to-mixed than purely negative, while TextBlob’s polarity-based approach classifies a much larger share of short, punchy political posts as neutral, reflecting its lower sensitivity to slang, capitalization emphasis,

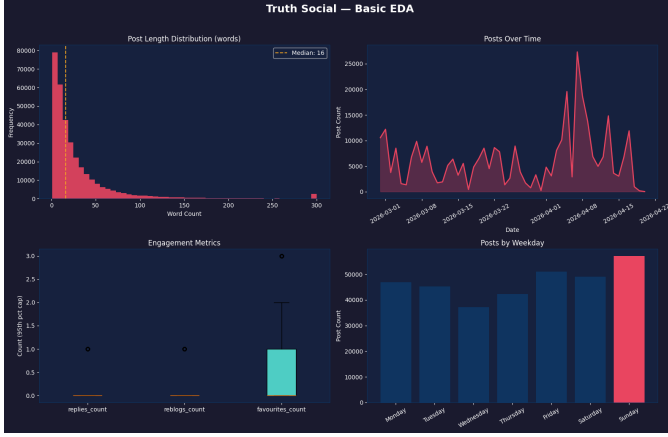


Fig. 1: Exploratory data analysis overview: post-length distribution, posting volume over time, engagement metric boxplots, and weekday posting frequency.

TABLE I: Sentiment Classification Comparison Across Models

Model	Positive	Neutral	Negative
VADER (full, $n=328,968$)	39.2%	19.0%	41.8%
TextBlob (full, $n=328,968$)	33.2%	43.4%	23.3%
RoBERTa (sample, $n=5,000$)	65.2%	10.7%	24.0%

and emoji-adjacent text relative to VADER and the Twitter-trained transformer. The VADER mean compound score across the full corpus was -0.028 , indicating a discourse that is, in aggregate, very close to neutral but tilted slightly negative.

Fig. 2 presents the full multi-model sentiment dashboard, including the VADER compound-score distribution (visibly bimodal, with mass concentrated near the extremes ± 1.0 as well as at exact neutrality, consistent with short exclamatory posts), the TextBlob polarity–subjectivity scatter, and the 3-day rolling average sentiment trend, which shows an initial sharply positive reaction that decays into a persistently negative regime as the conflict progressed, before partially recovering toward the end of the observation window.

C. Emotion Analysis

Of the 328,968 posts, 274,361 (83.4%) contained at least one word matched by the NRC lexicon. Table II shows the resulting dominant-emotion distribution. *Trust* is the single most frequent dominant emotion (28.5% of all posts), followed by *anger* (20.2%) and *fear* (15.5%), together indicating a discourse that mixes patriotic/supportive affect with substantial hostility and apprehension. Lower-intensity emotions such as *joy* (1.2%) and *disgust* (2.2%) are comparatively rare as dominant categories, although they remain present in blended form across many posts. Fig. 3 visualizes both the average per-post emotion intensity and the dominant-emotion share.

D. Keyword and N-gram Analysis

Table III lists the ten highest-scoring TF-IDF unigrams together with the most frequent bigrams and trigrams. Beyond substantive terms (“iran,” “war,” “regime,” “israel”), the n -gram analysis surfaces two notable non-substantive patterns:

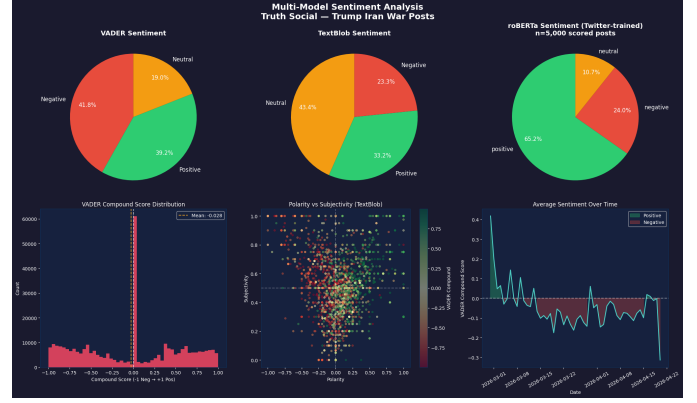


Fig. 2: Multi-model sentiment analysis dashboard: VADER, TextBlob, and RoBERTa label distributions; VADER compound-score histogram; polarity–subjectivity scatter; and rolling average sentiment over time.

TABLE II: Dominant Emotion Distribution (NRC Lexicon, $n=274,361$ matched posts)

Emotion	Posts	% of Corpus
Trust	93,901	28.5%
Anger	66,370	20.2%
Fear	50,984	15.5%
Anticipation	33,007	10.0%
Surprise	10,464	3.2%
Sadness	8,553	2.6%
Disgust	7,163	2.2%
Joy	3,919	1.2%

religiously-inflected patriotic phrasing (“god bless,” “bless america”) and meme-driven repetition (“taco taco,” “epstein epstein”), reflecting the platform’s characteristic mixture of sincere political commentary and in-group internet humor. Fig. 4 presents the corresponding horizontal bar charts.

Fig. 5 presents word clouds computed separately for the overall corpus and for each VADER-labeled sentiment subgroup, offering a qualitative complement to the quantitative n -gram counts.

E. Named Entity Recognition

Table IV summarizes entity type frequency over the 3,000-post NER sample, and Fig. 6 shows the top mentioned entities per type. **PERSON** and **GPE** (geo-political entity) mentions dominate, led by “Trump”/“Donald Trump” and “Iran”/“America”/“Israel,” respectively, confirming that the discourse is tightly centered on the named principal actors of the conflict. “Khamenei” is the most frequently mentioned non-U.S. political figure, and **NORP** entities (nationalities/religious/political groups) such as “MAGA” appear frequently, reflecting in-group identity signaling.

F. Topic Modeling

Table V lists the eight LDA topics with their top associated words and estimated document counts. The topics separate cleanly into: informal/profanity-laden commentary (Topic 0); regime-change and Iranian-people-focused discourse (Topic

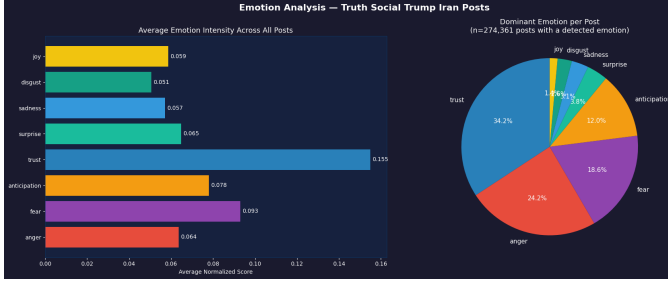


Fig. 3: Average NRC emotion intensity across all posts (left) and dominant-emotion share among posts with at least one lexicon match (right).

TABLE III: Top Keywords and N -grams

Top Unigrams (TF-IDF)	Top Bigrams (count)	Top Trigrams (count)
your (0.053)	god bless (15,538)	epstein epstein epstein (3,533)
iran (0.053)	iranian people (8,387)	taco taco taco (3,533)
them (0.041)	united states (8,318)	god bless america (3,387)
thank (0.040)	people iran (5,818)	your attention matter (1,218)
people (0.039)	thank mr (5,643)	thank your attention (1,218)
mr (0.033)	middle east (4,513)	united states america (1,218)
us (0.031)	american people (3,887)	bless god bless (1,241)
war (0.031)	bless america (3,819)	files epstein files (1,218)
god (0.030)	regime change (2,962)	epstein files epstein (1,218)
now (0.026)	strait hormuz (3,432)	god bless god (1,166)

1); military/nuclear escalation language (Topic 2); generic conversational text (Topic 3); diplomacy/deal-focused language referencing Israel (Topic 4); personal-pronoun-heavy commentary (Topic 5); war/oil/geopolitical-conspiracy language, notably co-occurring with “Epstein” (Topic 6); and religious/patriotic support language (Topic 7). Fig. 7 shows the corresponding word-weight bar charts.

As a semantic cross-check, BERTopic (fit on a 5,000-post sample) produced a larger set of finer-grained clusters; Fig. 8 shows the ten most populous, which include an oil/geopolitics-focused cluster referencing China and Venezuela, corroborating and refining the broader “war/oil” theme identified by LDA Topic 6.

G. Aspect-Based Sentiment Analysis

Table VI and Fig. 9 present the aspect-level sentiment breakdown across 390,104 total aspect-mention pairs (a single post may reference multiple aspects). **Trump Leadership** is both the most frequently mentioned aspect (104,626 mentions) and the only aspect with a clearly positive average sentiment (+0.115, 53.9% positive). **War/Conflict** shows the most negative average sentiment (−0.345, 70.5% negative posts), followed by **Media** (−0.225) and **Economy** (−0.189). **Diplomacy** is the only other aspect with net-positive sentiment (+0.058), suggesting that audience approval tracks negotiation and de-escalation framing even amid an overall hostile discourse toward the conflict itself.

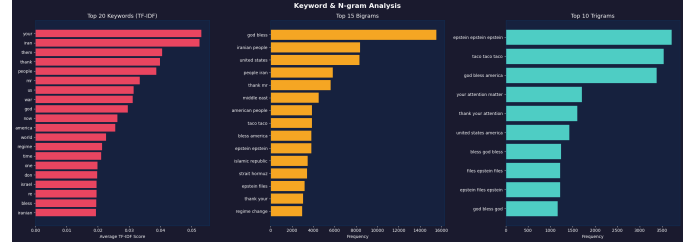


Fig. 4: Top 20 TF-IDF keywords, top 15 bigrams, and top 10 trigrams.

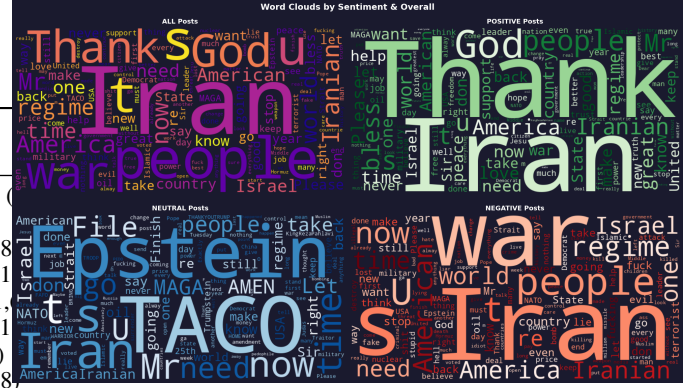


Fig. 5: Word clouds for the full corpus and for Positive, Neutral, and Negative sentiment subgroups.

H. Temporal Analysis

Fig. 10 tracks daily posting volume, 3-day rolling average sentiment, and the daily percentage breakdown of Positive/Neutral/Negative posts across the full February 28 – April 20, 2026 observation window. Volume shows several sharp spikes coinciding with major escalation and de-escalation announcements. Sentiment opens strongly positive, transitions into a sustained negative regime through the middle of the observation window as military operations intensified, and shows a partial late-window recovery, with a pronounced negative dip in the final days of the recorded data.

I. Engagement Analysis

Table VII reports mean total engagement (replies + reblogs + favourites) stratified by VADER sentiment label. Positively-toned posts receive roughly $3\times$ the average engagement of neutral posts and $1.7\times$ that of negatively-toned posts, indicating that affectively charged – and particularly positive/supportive – content is systematically more likely to be amplified. Fig. 11 presents the full correlation matrix among sentiment, subjectivity, word count, and engagement metrics, together with a sentiment-vs-engagement scatter plot.

J. Representative Discourse Examples

Manual inspection of the highest-engagement posts (all authored by the original poster rather than commenters) shows that the single most-engaged reaction (145,325 total engagement) was a strongly positive post celebrating the death of an Iranian leader as a form of justice; the second most-engaged

TABLE IV: Named Entity Mentions by Type (Sample, $n=3,000$ posts)

Type	Mentions	Top Entities
PERSON	1,811	Trump, God, Khamenei
GPE	1,745	Iran, America, Israel, US
ORG	1,114	Trump, MAGA, CCP
NORP	1,016	(nationalities/political groups)
LOC	110	—
LAW	40	—
EVENT	22	—

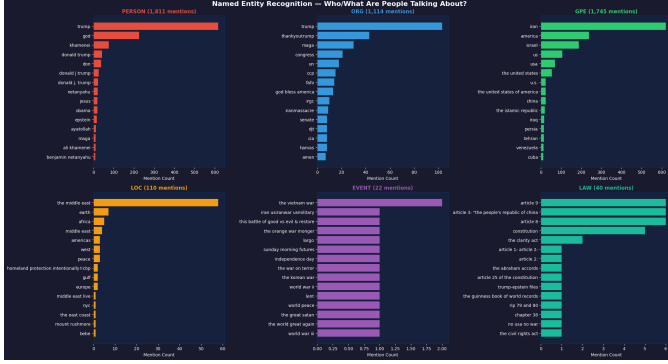


Fig. 6: Top mentioned named entities by type: PERSON, ORG, GPE, LOC, EVENT, LAW.

post (131,936) issued an ultimatum regarding the Strait of Hormuz and was scored negative due to its confrontational and profane phrasing despite its assertive tone; and a subsequent post announcing a two-week ceasefire suspension (99,180 engagement) was scored positive, illustrating that diplomacy-oriented announcements are received more favorably than escalatory ones, consistent with the aspect-level findings in Section IV-F. Consistent with the keyword analysis, the most extreme VADER-scored posts at both polarity ends were frequently short, mechanically repeated phrases (e.g., a single laudatory phrase or accusation repeated dozens of times), reflecting coordinated or emphatic commenting behavior rather than long-form argumentation.

V. DISCUSSION

The combined evidence across sentiment, emotion, and aspect-based analyses paints a picture of a discourse that is emotionally intense but not monolithically hostile or supportive. The co-dominance of *trust* and *anger* as the leading detected emotions (Table II), together with the aspect-level split between positive sentiment toward “Trump Leadership”/“Diplomacy” and negative sentiment toward “War/Conflict”/“Media” (Table VI), suggests that much of the audience’s negativity is directed at the conflict and its coverage rather than at the platform’s central figure. The divergence between VADER/RoBERTa and TextBlob sentiment distributions (Table I) further illustrates a well-known methodological caveat in social media NLP: lexicon models tuned for informal, capitalized, and punctuation-heavy text (VADER) or trained on in-domain Twitter data (RoBERTa) are more sensitive to the expressive conventions of this corpus than

TABLE V: LDA Topic Summary (8 Topics)

#	Top Words	Posts
0	mr, thank, pope, sir, trust, jesu	26,917
1	people, regime, world, iranian, iran, country	57,779
2	iran, military, nuclear, israel, destroy, irgc	30,228
3	your, them, don't, go, know, good, need	44,792
4	iran, must, deal, israel, finish, job	19,504
5	his, re, your, him, me, one, before	36,832
6	war, why, stop, oil, epstein, taco	80,796
7	god, bless, great, america, love, maga	32,120

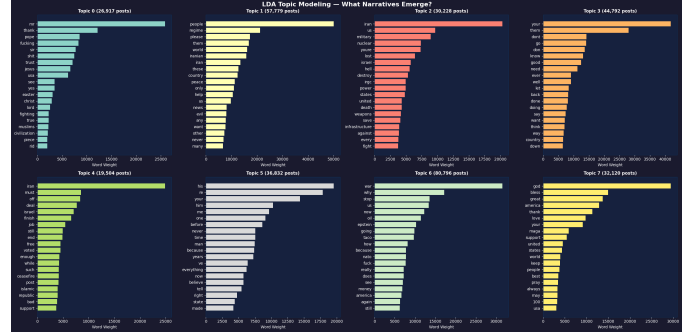


Fig. 7: LDA topic word-weight distributions for all 8 fitted topics.

a general-purpose polarity model (TextBlob), which defaults a much larger share of posts to neutral. The prominence of meme-driven n -grams (e.g., repeated “epstein,” “taco”) alongside substantive geopolitical vocabulary in both the TF-IDF and LDA results indicates that a non-trivial fraction of the discourse volume consists of in-group internet culture rather than deliberative political commentary, a pattern that should be accounted for when using volume-based metrics (e.g., raw mention counts) as a proxy for substantive public opinion.

VI. LIMITATIONS

Several limitations qualify these findings. First, the RoBERTa sentiment and BERTopic analyses were computed on non-random computational subsamples (the first 5,000 posts by row order) rather than the full corpus, which may not be representative of later time periods if discourse composition shifted over the observation window. Second, VADER, TextBlob, and the NRC lexicon are rule/lexicon-based and can be misled by sarcasm, coded language, and platform-specific slang not present in their training/lexicon resources. Third, Truth Social’s user base is not representative of the general public, so these findings characterize reaction within a specific partisan-leaning platform community and should not be generalized as a measure of broader public opinion. Finally, aspect keyword matching is a simple substring/keyword-based heuristic and may both over- and under-count aspect mentions relative to a fully disambiguated entity-linking approach.

VII. CONCLUSION

This report applied a comprehensive, multi-method NLP pipeline to 328,968 Truth Social posts and comments reacting to President Trump’s 2026 Iran conflict communications. Overall sentiment was found to be near-neutral with a slight

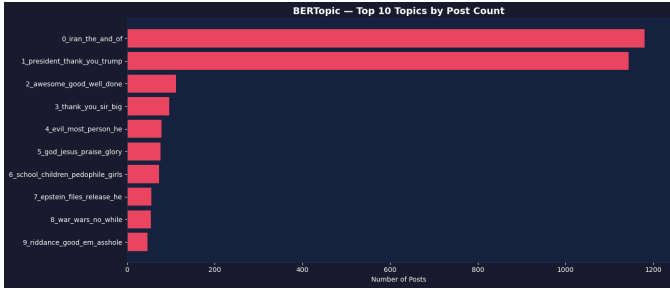


Fig. 8: BERTopic top 10 topics by post count (5,000-post sample).

TABLE VI: Aspect-Based Sentiment Summary

Aspect	Mentions	Avg. Sent.	%Pos	%Neg
Trump Leadership	104,626	+0.115	53.9%	36.3%
Iran Military	70,870	-0.063	42.1%	49.6%
War/Conflict	58,310	-0.345	26.1%	70.5%
National Security	28,914	-0.150	39.8%	58.5%
Diplomacy	23,649	+0.058	51.3%	44.3%
Israel	23,445	-0.174	35.2%	57.1%
Economy	23,360	-0.189	34.1%	56.4%
Biden/Democrats	21,262	-0.144	37.3%	55.3%
US Military	19,163	-0.047	44.8%	50.0%
Media	16,505	-0.225	31.6%	61.1%

negative lean, driven substantially by hostility toward the war itself and its media coverage rather than toward Trump’s leadership, which was viewed comparatively favorably. Emotionally, *trust* and *anger* dominate, and topic modeling reveals a discourse spanning military escalation, diplomacy, religious-patriotic identity expression, and meme culture. Engagement is systematically higher for positively-toned content, and sentiment shows clear temporal dynamics tied to the trajectory of the conflict. These results demonstrate the value of combining lexicon-based, transformer-based, and topic-modeling approaches to obtain a nuanced, multi-faceted picture of public reaction to a real-time geopolitical crisis on social media.

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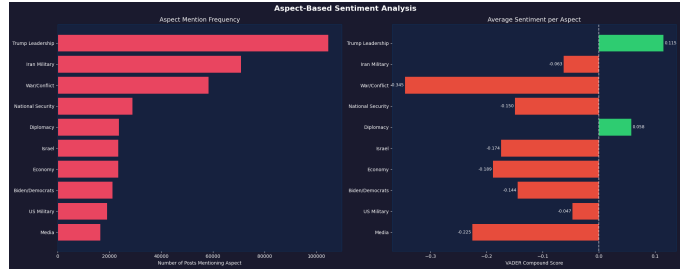


Fig. 9: Aspect mention frequency (left) and average VADER sentiment per aspect (right).

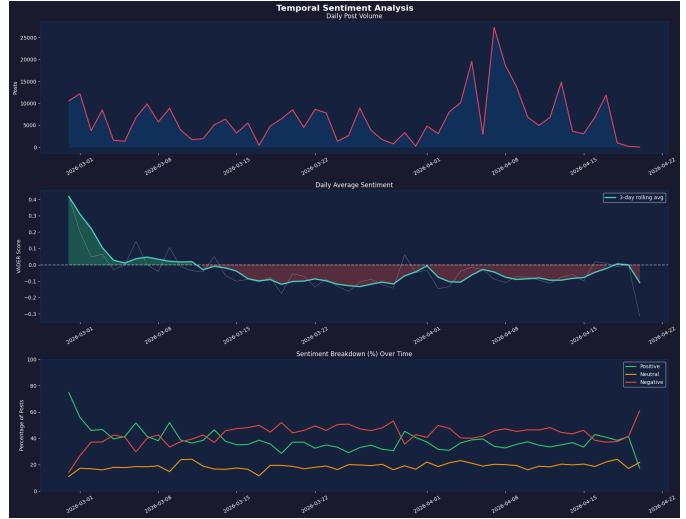


Fig. 10: Temporal analysis: daily post volume, 3-day rolling average sentiment, and daily sentiment-label percentage breakdown.

TABLE VII: Mean Total Engagement by VADER Sentiment Label

Sentiment Label	Mean Engagement
Positive	23.64
Negative	14.02
Neutral	7.82

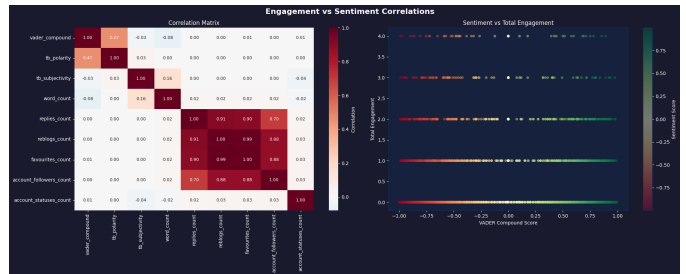


Fig. 11: Correlation matrix of sentiment, subjectivity, word count, and engagement metrics (left); sentiment vs. total engagement scatter plot (right).